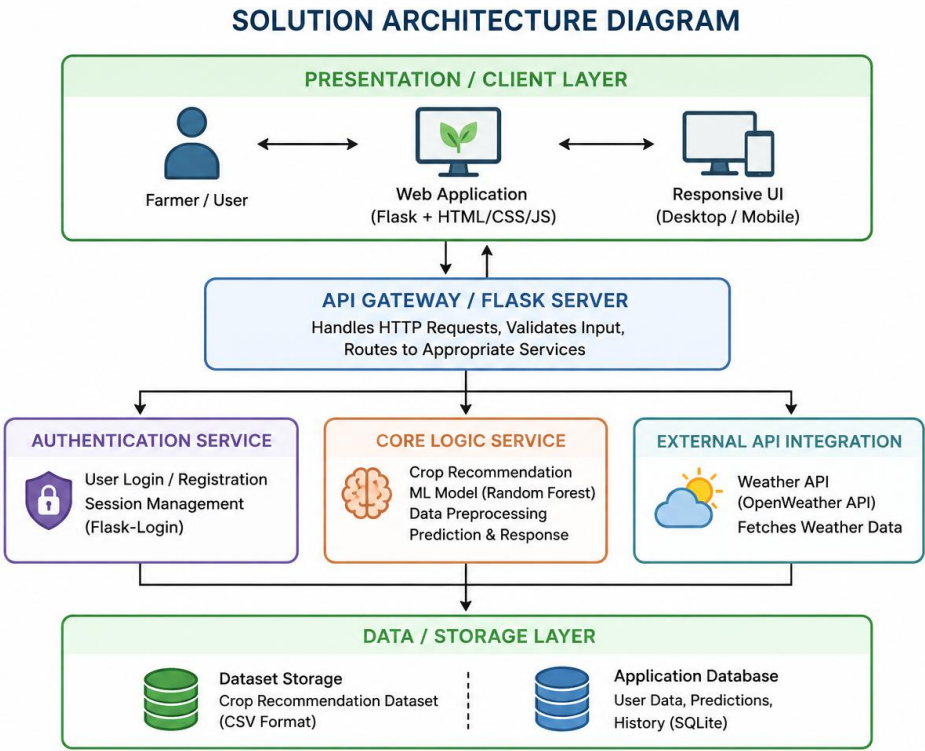


Solution Architecture

Field	Value
Date	05-07-2026
Team ID	(Mee Team ID)
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Solution Architecture Diagram



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a user-friendly interface where farmers enter soil and weather details and receive crop recommendations.	Flask, HTML, CSS, JavaScript
API Gateway	Handles user requests, validates input, and routes them to the prediction service.	Flask
Core Logic Service	Processes input data, executes the machine learning model, and predicts the most suitable crop.	Python, Scikit-learn, Pandas, NumPy
External API Integration	Retrieves weather-related information to improve crop recommendation accuracy.	OpenWeather API
Database / Storage	Stores datasets, user inputs, prediction history, and application data.	CSV Dataset, SQLite